

Capacity-Constrained Target Assignment and Recurrent Multi-Agent Reinforcement Learning for Cooperative Interception of Maneuvering UAV Swarms

Abstract—The swarm attack of unmanned aerial vehicles (UAVs) poses a significant threat to future battlefield operations. Achieving efficient and precise cooperative interception against attacking UAV swarms with highly dynamic maneuvering capabilities and adaptive penetration strategies poses a critical challenge for UAV interceptor swarms. This study addresses the cooperative interception of maneuvering targets in swarm attacks by decomposing the problem into two key tasks: target assignment and cooperative interception guidance. For target assignment, an Improved Dung Beetle Optimization (IDBO) algorithm is proposed to search directly over discrete assignments while enforcing target coverage and capacity. For cooperative interception guidance, a recurrent Multi-Agent Proximal Policy Optimization (R-MAPPO) controller is used under centralized training and decentralized execution. Its three-dimensional observation and reward explicitly represent relative altitude, line-of-sight geometry, group time-to-go dispersion, terminal interception, and control effort. A stored capacity-feasible assignment artifact is passed unchanged to the guidance environment, providing an auditable assignment-to-guidance interface rather than a regenerated end-to-end optimizer run. Representative three-dimensional engagements, a 30-seed assignment study, and separate software channel-impairment tests quantify both successful operation and a delayed-coordination failure. The results support feasibility under the tested conditions, while no claim of global optimizer convergence, Nash equilibrium, or hardware-in-the-loop validation is made.

Note to Practitioners—This paper is motivated by practical UAV swarm defense scenarios, where multiple interceptors must cooperatively engage maneuvering aerial targets under communication and onboard maneuver constraints. The revised framework exposes the interface between target assignment and guidance: a stored capacity-feasible assignment artifact is consumed unchanged by a recurrent MAPPO controller. The simulation now uses three-dimensional fixed-wing kinematics and reports representative trajectories together with separate tests of sensing delay, noise, command lag, and packet loss. These are mathematical and software channel simulations, not flight or hardware-in-the-loop experiments. The observed delayed-coordination case also shows that communication impairment can preserve individual interception while breaking group simultaneity; operational use therefore requires online health monitoring, reassignment, and hardware validation beyond the present study.

Index Terms—Cooperative interception, deep reinforcement learning, centralized training with decentralized execution, target assignment, UAV swarm.

I. INTRODUCTION

AS warfare evolves toward intelligent swarm operations, the likelihood of encountering attacks from adversarial

UAV swarms significantly increases during combat. Consequently, the efficient interception of incoming swarm targets becomes a critical imperative for future battlefield operations. To address the challenge of cooperative interception against maneuvering attacking swarms with adaptive penetration strategies, there is an urgent need to advance academic research in swarm target interception technologies. This study aims to enhance interceptors with intelligent cooperative interception capabilities, enabling effective navigation of complex challenges in modern battlefield environments.

Multi-target cooperative interception encompasses two key tasks: target assignment and cooperative interception. The effectiveness of multi-target interception hinges on the optimality of target assignment, establishing it as a primary focus in swarm target interception strategies. As the number of UAVs increases, the complexity of target assignment escalates significantly, resulting in a multi-constrained, multi-parameter optimization problem. The objective is to develop a rapid and efficient target assignment method for allocating specific attacking targets to interceptors in operational scenarios [1]. The target assignment problem involves complex constraints, and the non-dominated sorting algorithm can dynamically generate a set of reference points with good convergence and distribution performance [2]–[4]. Moreover, employing an on-line operator selection mechanism can enhance the efficiency of the algorithm [5], [6]. The target assignment problem can be solved using multi-information particle swarm optimization to find the solution that yields the highest gain [7], [8]. Introducing information from surrounding particles can enhance the stability of the traditional particle swarm algorithm while also addressing the issue of local convergence in the target assignment problem [9]. The genetic algorithm is a common approach for solving the target assignment problem [10]–[13], and a GA with a novel crossover operator can further enhance algorithmic efficiency [14]. Ant colony optimization is also an effective strategy for solving the target assignment problem [15], and its combination with a greedy algorithm can improve the accuracy of parallel computation [16]. Under multiple constraints, researchers have proposed various heuristic algorithms to solve the target assignment problem, such as neural networks [15], auction algorithm [17], tabu search [18], simulated annealing algorithm [19], VLSN [20]. However, in cooperative interception scenarios, the aforementioned algorithms struggle to handle the highly dynamic nature of the battlefield due to their high time complexity, and they fail to meet the dual requirements of both speed and optimality.

The effectiveness of the terminal guidance phase in multi-target interception is significantly influenced by the cooper-

ative guidance law, a critical factor requiring thorough consideration. The target assignment strategy transforms swarm operations into a coordinated multi-to-one interception problem, with current research focusing on three key areas: angular coordination, temporal coordination, and intelligent cooperation. Significant progress has been made in temporal coordination. Classical proportional navigation-based guidance laws enable synchronized multi-missile attacks on a single target [21]. Adaptive control methods adjust attack timing based on real-time operational conditions, making them suitable for intercepting maneuvering targets [22], [23]. Sliding Mode Control (SMC) is widely adopted for cooperative interception [24], with SMC incorporating miss distance and attack time constraints enhancing interception accuracy [25]–[27]. Considerable research has also been conducted on angular coordination. From a traditional control perspective, backstepping control [28] and leader-follower cooperative methods [29] significantly improve interception efficiency. SMC with angular constraints is prevalent in cooperative interception, encompassing approaches based on impact angle constraints [30] and line-of-sight angle constraints [31]. Optimal control methods applied to impact angle-constrained cooperative guidance enhance precision by integrating impact angle and line-of-sight angle constraints [32], [33].

Recent advancements in intelligent technologies have driven extensive research into cooperative interception strategies for UAV swarms, with intelligent algorithms demonstrating substantial potential in addressing complex multi-target scenarios [34], [35]. Optimization-based approaches, such as Particle Swarm Optimization (PSO), excel in collaborative attack problems by autonomously optimizing resource allocation and maximizing operational efficiency [35], [36]. Search-based algorithms, like A* [37], enable efficient path planning for coordinated attacks. Control-based methods, including Model Predictive Control (MPC) [38], provide robust guidance under dynamic constraints. Moreover, the integration of neural networks with reinforcement learning, particularly deep reinforcement learning (DRL), enhances the adaptability and intelligence of interception strategies [39]–[41]. DRL enables UAV swarms to learn complex policies from high-dimensional data, optimizing target assignment and maneuvering in real-time to counter maneuvering targets [42]. In contrast, conventional cooperative interception guidance methods rely heavily on linearization and simplifying assumptions to design feasible guidance laws [43]. These methods often struggle to address the complexity of dynamic, non-linear environments involving maneuvering swarm targets. Data-driven intelligent cooperative interception methods, by contrast, leverage extensive datasets to learn adaptive strategies, significantly reducing dependence on restrictive assumptions [44]. By modeling complex interactions through neural networks and DRL, these methods achieve greater robustness and flexibility. Despite advancements, intelligent cooperative interception methods struggle with maneuvering swarm targets in complex dynamic environments due to limited model generalization, low training efficiency, and low interception success rates, particularly in large-scale swarm combat requiring precise temporal coordination [45]. Addressing these challenges requires further research

into scalable, generalizable, and efficient algorithms to fully realize the potential of intelligent cooperative interception.

In summary, existing cooperative interception methods for countering attacking UAV swarms with adaptive penetration capabilities suffer from low interception success rates in complex dynamic environments. Specifically, target assignment methods exhibit low allocation efficiency, while cooperative interception guidance strategies struggle with terminal overload saturation and insufficient adaptability to dynamic target maneuvers in practical scenarios, hindering effective swarm coordination. Consequently, more research is necessary to address these challenges.

This study proposes a swarm cooperative interception method based on multi-agent reinforcement learning and IDBO algorithm, enhancing target assignment efficiency and guidance strategy performance to achieve precise and efficient interception of maneuvering targets. The proposed intelligent cooperative interception algorithm offers the following advantages.

- 1) A capacity-constrained IDBO assignment method is formulated for the many-to-one interception problem. Its discrete rolling, dancing, breeding, and stealing updates are combined with feasibility repair and elitist acceptance. A 30-seed coefficient study and a bounded-delay information-propagation experiment characterize empirical stability, scale, and communication cost without claiming global or game-theoretic convergence.
- 2) A recurrent MAPPO guidance controller is specified consistently with the deployed checkpoints. A 17-dimensional local observation captures three-dimensional relative geometry, closing motion, group time-to-go dispersion, and previous commands; a GRU provides temporal memory, and the actor outputs bounded axial, yaw-normal, and pitch-normal loads.
- 3) The assignment-to-guidance data interface is audited for 20 defenders and eight targets using a stored capacity-feasible artifact; this is not presented as a regenerated IDBO-to-guidance run. The revised evaluation reports representative three-dimensional R-MAPPO and capacity-matched PN engagements, plus separate delay/noise/lag and software channel-impairment tests. Reward shaping and PPO clipping are treated as standard task-adaptation and training components rather than standalone methodological innovations.

The paper is structured as follows: Section II presents relevant problem modeling and foundational theory. Section III and Section IV provide a detailed description of our approach, including its framework, modules, algorithms, and rationale. Section V covers experiments, results, and discussions. Lastly, Section VI summarizes the paper and outlines future directions.

II. BACKGROUNDS AND PRELIMINARIES

In this section, we first introduce the modeling of multi-target interception of UAVs, and then review reinforcement learning methods.

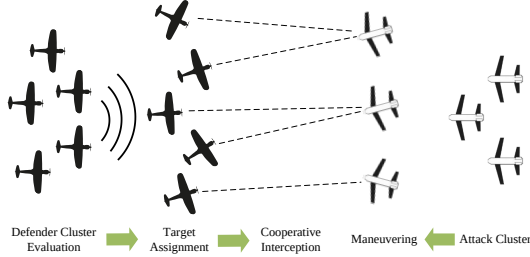


Fig. 1. Cluster target interception problem.

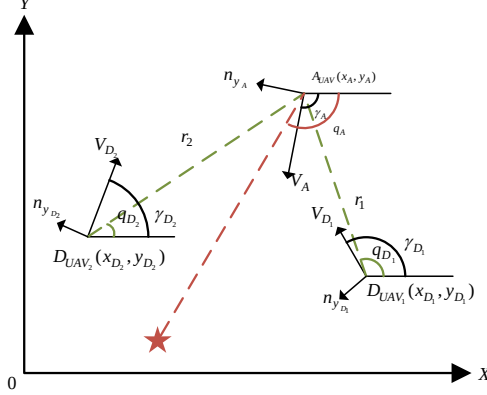


Fig. 2. Schematic diagram of confrontation and interception.

A. Problem Modeling

The multi-target intelligent interception problem can be decomposed into two subproblems: target assignment and intelligent cooperative interception, with the swarm combat scenario illustrated in Fig. 1. In this scenario, the attacking swarm aims to penetrate a high-value target area, while the defending swarm seeks to fully intercept the incoming UAVs. During the target assignment phase, the defending swarm assesses its own flight status and that of the attacking swarm using onboard sensors, subsequently allocating each defender to an appropriate interception target via the target assignment strategy. In the intelligent cooperative interception phase, defenders assigned to the same target employ coordinated maneuvering strategies to intercept and neutralize the target.

The object of study is a fixed-wing UAV. The horizontal engagement geometry in Fig. 2 is retained to define range and bearing, while the revised simulation and guidance model propagate the full three-dimensional position, pitch, and yaw states.

The three-dimensional point-mass kinematic model is

$$\begin{cases} \dot{x} = V \cos \vartheta \cos \psi, & \dot{y} = V \cos \vartheta \sin \psi, \\ \dot{z} = V \sin \vartheta, & \dot{V} = gn_x, \\ \dot{\psi} = \frac{gn_y}{V \max(\cos \vartheta, 0.25)}, & \dot{\vartheta} = \frac{gn_z}{V}, \end{cases} \quad (1)$$

where $\mathbf{p} = [x, y, z]^T$, V , ψ , and ϑ denote position, speed, yaw, and pitch. The normalized commands n_x , n_y , and n_z are axial, yaw-normal, and pitch-normal loads, respectively.

In an interception mission, the relative motion relationship between the attacking and defending UAVs can be described as:

$$\begin{cases} \dot{r} = V_A \cos(\gamma_A - q) - V_D \cos(\gamma_D - q) \\ r\dot{q} = V_A \sin(\gamma_A - q) - V_D \sin(\gamma_D - q) \end{cases} \quad (2)$$

where r is the inter-UAV distance, q is the line-of-sight angle, V_A (V_D) is the velocity of the attacking (defending) UAV, and γ_A (γ_D) is the corresponding heading angle.

For the horizontal special case $\vartheta = n_z = 0$, (1) reduces to the two-dimensional relative-motion relations in (2). This reduction is used only for defining the assignment geometry; all revised case simulations integrate (1).

During cooperative interception, the remaining impact time is defined as:

$$t_{go} = \frac{r}{|\dot{r}|} \quad (3)$$

The Zero-Effort Miss (ZEM) is defined as:

$$ZEM = \frac{r^2 \cdot \dot{q}}{|\dot{r}|} \quad (4)$$

B. Reinforcement Learning

Reinforcement learning (RL) is built on the Markov Decision Process (MDP), comprising state space $\mathcal{S}$, action space $\mathcal{A}$, reward function R , policy π , and state transition function P . At each step, the agent selects action a from state s , receives reward r , and transitions to s' . The key value functions are:

$$V^\pi(s) = E_\pi \left[\sum_{t=0}^{\infty} \gamma_{RL}^t r_t \mid s_0 = s \right] \quad (5)$$

$$Q^\pi(s, a) = E_\pi \left[\sum_{t=0}^{\infty} \gamma_{RL}^t r_t \mid s_0 = s, a_0 = a \right] \quad (6)$$

where $\gamma_{RL} \in [0, 1]$ is the discount factor.

III. TARGET ASSIGNMENT VIA CAPACITY-CONSTRAINED IDBO

This section formulates cooperative target assignment as a weighted bipartite matching problem and solves it with a capacity-constrained Improved Dung Beetle Optimization (IDBO) algorithm.

A. Problem Formulation

Let $\mathcal{U} = \{u_1, u_2, \dots, u_M\}$ and $\mathcal{T} = \{t_1, t_2, \dots, t_N\}$ denote the interceptor and target sets, respectively. Figure 3 illustrates the bipartite assignment structure.

The binary decision variable $X_{ij} = 1$ if UAV u_i is assigned to target t_j and 0 otherwise. The edge weight P_{ij} quantifies the interception probability. Intuitively, a desirable interceptor-target edge should have both a small predicted miss tendency and a favorable heading geometry; therefore, the weight combines ZEM-based feasibility and angular advantage:

$$P_{ij} = w_{ZEM} P_{ZEM}(i, j) + w_\sigma P_\sigma(i, j) \quad (7)$$

where $w_{ZEM} + w_\sigma = 1$ and both weights are nonnegative. The Zero-Effort Miss (ZEM) component is:

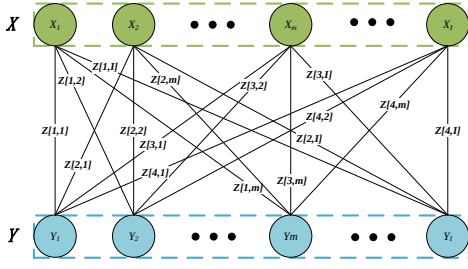


Fig. 3. Schematic diagram of weighted dichotomy.

$$P_{\text{ZEM}}(i, j) = \exp \left[-\frac{1}{2} \left(\frac{\text{ZEM}(i, j)}{\overline{\text{ZEM}}} \right)^2 \right] \quad (8)$$

where $\overline{\text{ZEM}} = \frac{1}{MN} \sum_{i=1}^M \sum_{j=1}^N \text{ZEM}(i, j)$.

The angular advantage component is:

$$P_{\sigma}(i, j) = \exp \left[-\frac{1}{2} \left(\frac{S_{\sigma}(i, j)}{\overline{S_{\sigma}}} \right)^2 \right] \quad (9)$$

where $\overline{S_{\sigma}} = \frac{1}{MN} \sum_{i=1}^M \sum_{j=1}^N S_{\sigma}(i, j)$, with $S_{\sigma}(i, j) = |\gamma_{ij} - q_{ij}|$ denoting the heading error. The assignment objective below converts the tactical goal of “neutralize as many attackers as possible” into a survival-probability minimization problem. Under many-to-one assignment, each additional capable interceptor assigned to the same target multiplies that target’s survival probability by another failure term:

$$\begin{aligned} \min_{\mathbf{x}} \quad & J = \sum_{j=1}^N \left[\prod_{i=1}^M (1 - P_{ij})^{X_{ij}} \right] \\ \text{s.t.} \quad & \sum_{j=1}^N X_{ij} = 1, \quad \forall i \in \{1, \dots, M\} \\ & \sum_{i=1}^M X_{ij} \geq 1, \quad \forall j \in \{1, \dots, N\} \\ & \sum_{i=1}^M X_{ij} \leq L_{\max}, \quad \forall j \in \{1, \dots, N\} \\ & X_{ij} \in \{0, 1\}, \quad \forall i, j \end{aligned} \quad (10)$$

where $\prod_{i=1}^M (1 - P_{ij})^{X_{ij}}$ yields the survival probability of t_j , which decreases geometrically with the number of assigned interceptors.

B. Capacity-Constrained IDBO and Input Dissemination

The optimizer uses a discrete assignment vector $\mathbf{x} = [x_1, \dots, x_M]^T$, where $x_i = j$ assigns interceptor u_i to target t_j . This encoding implements the row constraint in (10) directly. A deterministic repair operator $\mathcal{R}$ moves the least-cost interceptor whenever a target is empty or contains more than $L_{\max}$ members; consequently, every evaluated candidate satisfies all three capacity constraints.

At iteration k , the population is divided equally among the four DBO roles. The discrete updates have the following operational interpretation: rolling copies selected components

Algorithm 1 Capacity-Constrained IDBO and Assignment Export

Require: Probability matrix $\mathbf{P}$, population size P , iteration budget K , capacity $L_{\max}$

Ensure: Feasible assignment vector $\mathbf{x}^*$

- 1: Initialize P random assignments and repair each with $\mathcal{R}$
- 2: Evaluate the target-survival objective in Eq. (10)
- 3: **for** $k = 0, \dots, K - 1$ **do**
- 4: Compute s_k and m_k from Eq. (11)
- 5: Rank the population and retain the best and second-best assignments
- 6: **for** each rolling, dancing, breeding, or stealing candidate **do**
- 7: Modify m_k assignment components using the role-specific rule
- 8: Apply capacity repair $\mathcal{R}$
- 9: Accept the candidate only if its cost does not increase
- 10: **end for**
- 11: Retain the best-so-far feasible assignment
- 12: **end for**
- 13: Export $\mathbf{x}^*$ and use it unchanged in the guidance environment

from the current best solution while retaining occasional random moves; dancing randomly redirects selected components; breeding recombines the best and second-best assignments; and stealing copies selected components from the best assignment. The number of modified components is controlled by

$$s_k = 0.32 \left(1 - \frac{k}{K-1} \right) + 0.04, \quad (11)$$

$$m_k = \max(1, \text{round}(s_k M)). \quad (12)$$

Thus, the search moves from broader exploration toward local exploitation but never becomes deterministic. Every candidate is repaired and accepted only if its objective in (10) does not exceed that of its parent. The best-so-far value is also retained. These two explicit implementation rules replace the previously ambiguous continuous preference equations.

Separately from the optimizer, we evaluate dissemination of fixed-snapshot input sources over a connected sparse graph. Nodes repeatedly forward the source identifiers they know, and delayed or dropped messages are retransmitted in subsequent rounds. This experiment measures only how long immutable information needs to reach all nodes. It does not execute distributed IDBO, assignment consensus, or failure-triggered reassignment.

The complete reproducible procedure is summarized in Algorithm 1. The assignment exported to guidance is a length- M integer vector rather than a separately reconstructed topology.

C. Convergence Scope, Complexity, and Delay Model

Elitist acceptance gives the limited but rigorous property

$$J_{\text{best}}^{(k+1)} \leq J_{\text{best}}^{(k)}, \quad J_{\text{best}}^{(k)} \geq 0. \quad (13)$$

Hence the recorded incumbent sequence converges to a finite limit. This does *not* prove that the population reaches consensus, that the limit is globally optimal, or that the assignment is a Nash equilibrium. The problem is nonconvex and discrete, and the stochastic search can terminate at a local plateau. We therefore report paired-seed empirical cost and stabilization statistics rather than an equilibrium claim.

For a candidate evaluation and repair implemented by direct scanning, the worst-case work is $O(MN)$. A population of size P and K iterations therefore costs $O(KPMN)$; memory

is $O(PM+MN)$. These are upper bounds for the implementation, not hardware-independent runtime guarantees. The separate input-source propagation experiment records synchronous message rounds. On a connected graph with bounded link delay, repeated delivery eventually disseminates immutable source information; this observation is not a convergence result for IDBO. With changing data, permanent link failure, or a disconnected graph, even this limited conclusion no longer applies.

IV. RECURRENT MAPPO FOR THREE-DIMENSIONAL COOPERATIVE INTERCEPTION

The revised guidance description is restricted to the architecture represented by the evaluated checkpoints. It is a recurrent MAPPO controller under centralized training with decentralized execution (CTDE). In particular, the deployed actor contains an MLP encoder, a GRU, and a Gaussian action head. The earlier trust-modulated exploration and attention-residual claims are not retained because they were not independently supported by the evaluated checkpoint or an ablation.

A. Actor–Critic Architecture and PPO Update

For interceptor i , the encoder and recurrent state update are

$$\mathbf{e}_{i,t} = f_{\text{MLP}}(\mathbf{o}_{i,t}; \theta_e), \quad (14)$$

$$\mathbf{h}_{i,t} = \text{GRU}(\mathbf{e}_{i,t}, \mathbf{h}_{i,t-1}; \theta_h). \quad (15)$$

The actor defines a diagonal Gaussian and then clips the sampled command to the flight envelope:

$$\begin{aligned} \pi_\theta(\mathbf{a}_{i,t} \mid \mathbf{o}_{i,t}, \mathbf{h}_{i,t-1}) &= \mathcal{N}(\boldsymbol{\mu}_\theta(\mathbf{h}_{i,t}), \text{diag}(\boldsymbol{\sigma}_\theta^2)), \\ \mathbf{a}_{i,t} &= [n_{x,i}, n_{y,i}, n_{z,i}]^\top, \\ \mathcal{A} &= [-0.1, 1] \times [-1, 1] \times [-1, 1]. \end{aligned} \quad (16)$$

The centralized critic receives the joint state during training, whereas each actor uses only its local observation and recurrent state during execution:

$$s_t = [\mathbf{o}_{1,t} \parallel \dots \parallel \mathbf{o}_{M,t}], \quad V_\phi(s_t) \in \mathbb{R}. \quad (17)$$

Generalized advantage estimation is

$$\delta_t = r_t + \gamma_{\text{RL}} V_\phi(s_{t+1}) - V_\phi(s_t), \quad (18)$$

$$\hat{A}_t = \sum_{\ell=0}^{T-t-1} (\gamma_{\text{RL}} \lambda_{\text{GAE}})^\ell \delta_{t+\ell}. \quad (19)$$

The actor uses the standard clipped PPO surrogate

$$L^{\text{clip}}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right], \quad (20)$$

where $r_t(\theta) = \pi_\theta(\mathbf{a}_t \mid \mathbf{o}_t) / \pi_{\theta_{\text{old}}}(\mathbf{a}_t \mid \mathbf{o}_t)$. Clipping regularizes sampled policy updates but does not impose a hard bound on the realized probability ratio and does not guarantee global convergence.

B. Three-Dimensional Observation and Reward

The 17-dimensional local observation is

$$\begin{aligned} \mathbf{o}_{i,t} &= [r/2000, \Delta z/1000, \dot{r}/100, e_\angle/\pi, V_i/65, V_T/65, \\ &\quad \vartheta_i/\pi, \psi_i/\pi, \ell_{iT}^\top, (t_{go,i} - \bar{t}_{go})/50, \\ &\quad \Delta t_{go}/50, (t_{go,i} - t_{go,\min})/50, n_{x,i}^-, n_{y,i}^-, n_{z,i}^-]^\top, \end{aligned} \quad (21)$$

where $\ell_{iT} \in \mathbb{R}^3$ is the unit line-of-sight vector, $e_\angle$ is the angle between the interceptor velocity and the LOS, $\Delta t_{go} = t_{go,\max} - t_{go,\min}$ within the assigned group, and the superscript $-$ denotes the previous applied load. The observation uses relative quantities and normalized scales so that the same actor can execute on homogeneous interceptors.

The implemented reward has six task terms:

$$r_{i,t} = \sum_{q \in \{d,a,h,c,s,e\}} w_q r_{i,t}^q, \quad (22)$$

$$r^d = \exp(-\alpha_d r), \quad r^a = -\alpha_a e_\angle^{p_a}, \quad (23)$$

$$r^h = R_{\text{hit}} \mathbb{I}(r < d_{\text{hit}}), \quad r^c = -\alpha_c |t_{go,i} - \bar{t}_{go}|^{p_c} + B_c, \quad (24)$$

$$r^s = -\alpha_s (\Delta t_{go})^{p_s} + B_s, \quad r^e = -\alpha_e \|\mathbf{a}_{i,t}\|_2^2 \Delta t. \quad (25)$$

The bonuses B_c and B_s are applied only inside their respective coordination tolerances. The nominal implementation uses

$$\begin{aligned} (w_d, w_a, w_h, w_c, w_s, w_e) &= (0.1, 1, 1, 1, 1, 1), \\ (\alpha_d, \alpha_a, \alpha_c) &= (10^{-3}, 10^{-2}, 5 \times 10^{-3}), \\ (\alpha_s, \alpha_e) &= (2 \times 10^{-2}, 5 \times 10^{-2}), \\ R_{\text{hit}} &= 3, \quad \tau_{\text{sync}} = 0.5 \text{ s}. \end{aligned}$$

C. Sensitivity and Convergence Scope

Reward sensitivity is best interpreted through invariant mission metrics rather than returns, because changing a reward weight changes the numerical definition of return. Increasing w_d favors early range reduction; increasing w_a favors velocity–LOS alignment; increasing w_c or w_s prioritizes within-group timing; and increasing w_e suppresses load usage. Excessive coordination weight can delay a geometrically favorable interceptor, whereas excessive energy weight can prevent the maneuver needed for a hit. Accordingly, the evaluation reports hit count, group arrival spread, miss distance, and load limits rather than comparing returns across reward definitions. A full multi-seed retraining sweep of these weights is not claimed by the present evidence; this limitation is stated explicitly to avoid interpreting an uncontrolled one-seed preview as sensitivity evidence.

R-MAPPO and conventional MAPPO share the same PPO-style nonconvex optimization and therefore the same lack of a global or Nash-convergence guarantee. The GRU changes the policy class by retaining observation history, which is useful under partial observation, but it does not strengthen the theoretical convergence guarantee. PPO clipping is a training-time operation and adds no actor operations during deployment. The deployed actor contains 533,800 parameters; inference uses only the MLP, GRU, and Gaussian mean head. Processor-specific latency was not measured, so no onboard deadline claim is made.

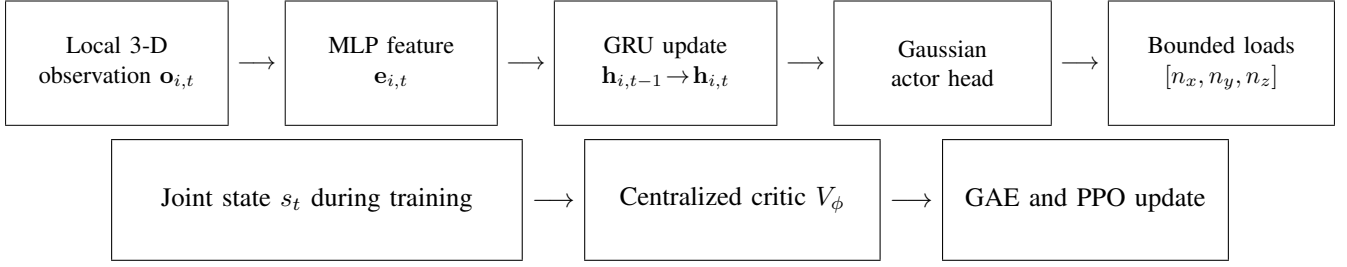


Fig. 4. Revised R-MAPPO data flow. The actor path shows one unambiguous temporal transition from $\mathbf{h}_{i,t-1}$ to $\mathbf{h}_{i,t}$; the critic and optimization path is used only during centralized training.

Algorithm 2 R-MAPPO Cooperative Interception Guidance

Require: Exported assignment $\mathbf{x}^*$, 3-D environment, horizon T

Ensure: Decentralized recurrent actor π_θ

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1: Form target groups directly from  $\mathbf{x}^*$ 
2: for each training episode do
3:   Reset 3-D states and recurrent states
4:   for  $t = 0, \dots, T - 1$  do
5:     for each interceptor  $i$  do
6:       Build  $\mathbf{o}_{i,t}$  from Eq. (21)
7:       Update  $\mathbf{h}_{i,t}$  with Eq. (15)
8:       Sample and clip  $[n_x, n_y, n_z]$  using Eq. (16)
9:     end for
10:    Propagate 3-D dynamics and compute Eq. (22)
11:    Store local observations, joint state, actions, rewards, and masks
12:  end for
13:  Compute GAE and update actor/critic with clipped PPO
14: end for
15: During deployment, execute each actor from local observation and GRU state
  
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D. Evaluation Criteria

For each target group $\mathcal{G}_j$, the terminal synchronization spread and miss distance are

$$E_{\text{sync},j} = \max_{i \in \mathcal{G}_j} t_{f,i} - \min_{i \in \mathcal{G}_j} t_{f,i}, \quad (26)$$

$$E_{\text{miss}} = \frac{1}{M} \sum_{i=1}^M \|\mathbf{p}_i(t_{f,i}) - \mathbf{p}_{T(i)}(t_{f,i})\|_2. \quad (27)$$

We additionally report the number of interceptor hits, the number of covered and synchronized target groups, and the maximum absolute command in each of the three load channels. A group is synchronized in the final case study when $E_{\text{sync},j} \leq 0.5$ s.

V. SIMULATION VERIFICATION

A. Revised Three-Dimensional Setup

The revised evaluation uses 20 defending UAVs and eight attacking UAVs with the three-dimensional dynamics in (1). The simulation time step is 0.05 s and the maximum horizon is 1500 steps (75 s), correcting the previously reported 25-step value, which belonged to an unrelated auxiliary benchmark. Table I gives the settings used by the saved case presets.

These explicit radial intervals replace the ambiguous D_1, D_2, D_3 notation and the inconsistent two-dimensional initialization sketch in the original manuscript. The two attacker profiles are: Case 1, a piecewise evasive maneuver; and Case 2, continuous sinusoidal weaving. Both include vertical motion. The saved trajectories start at the audited states in Table I and use identical initial horizontal states for R-MAPPO and PN.

TABLE I
REVISED THREE-DIMENSIONAL EVALUATION SETTINGS

Setting	Value
Defenders / attackers	20 / 8
Time step / maximum horizon	0.05 s / 1500 steps
Initial defender radius	52.63–65.44 m
Initial attacker radius	1987.72–2145.95 m
Initial defender / attacker altitude	0 m / 120 m
Initial defender speed	20.32–30.02 m/s
Initial attacker speed	30.23–45.87 m/s
Hit radius / synchronization threshold	3 m / 0.5 s
Defender load limits	$n_x \in [-0.1, 1], n_y, n_z \in [-1, 1]$

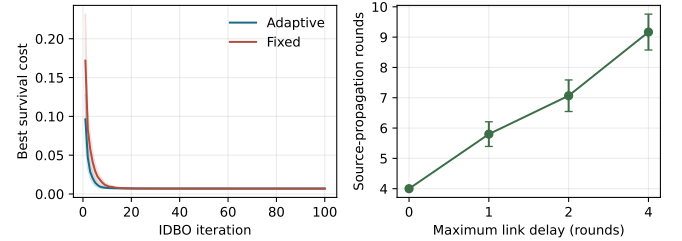


Fig. 5. Capacity-constrained IDBO study. Left: best target-survival cost for a common decaying mutation strength and a fixed strength (mean and one standard deviation over 30 seeds). Right: rounds required for fixed-source information propagation with 1% packet dropout; this is not an IDBO convergence curve.

TABLE II
IDBO COEFFICIENT-SCHEDULE STUDY (30 PAIRED SEEDS)

Schedule	Final cost	Median iteration to 1% band
Decaying 0.36 $\rightarrow$ 0.04	0.006975 $\pm$ 0.000141	19.0
Fixed 0.18	0.006942 $\pm$ 0.000083	21.5

B. IDBO Ablation, Weight Sensitivity, and Scale

The revised assignment study implements the 20×8 many-to-one decision in (10), rather than the inverse eight-variable surrogate used by the previous figures. Each variant uses 30 paired seeds, a population of 40, and 100 iterations. All 60 final assignments in the coefficient comparison satisfy the one-to-three interceptor capacity constraint.

The decaying schedule reaches its own 1%-of-final band slightly earlier, but it does not improve final cost: among the 30 paired runs it is better, equal, and worse in 1, 25, and 4

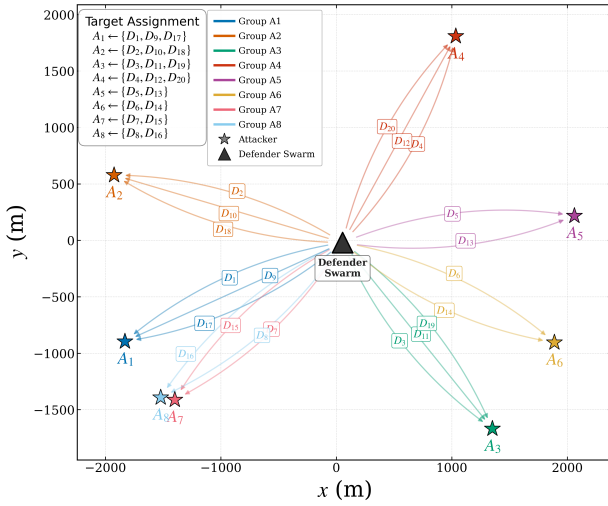


Fig. 6. Stored 20-to-8 capacity-feasible assignment passed unchanged to both three-dimensional guidance cases.

runs, respectively. We therefore do not claim a final-quality advantage from coefficient adaptation. Likewise, varying the ZEM weight $w_{ZEM} \in \{0.3, 0.5, 0.7\}$, with $w_\sigma = 1 - w_{ZEM}$, changes the optimized objective mean to 0.004611, 0.006986, and 0.009956. These absolute values are not ranked because the objective itself changes with the weight; the sweep demonstrates the expected tactical tradeoff rather than a universal best weight.

For 60 iterations and population 30, the final archived audit recorded mean runtimes of 0.417, 0.732, and 1.846 s for $M/N = 20/8, 40/16, 80/32$, respectively. Repeated audit executions showed that the absolute wall-clock values shift substantially with shared-host load; accordingly, only the increasing scale trend and the analytical $O(KPMN)$ implementation bound are retained. These timings are not evidence of an onboard deadline. In the separate 20-node input-propagation test, the mean rounds at 1% dropout were 4.00, 5.80, 7.07, and 9.17 for maximum link delays of 0, 1, 2, and 4 rounds. Removing one node still allowed the 19-node connected subgraph to disseminate all remaining source identifiers in 7.17 ± 0.46 rounds, but no task reassignment or guidance after vehicle loss was evaluated.

C. Assignment-to-Guidance Interface

Figure 6 shows the stored assignment artifact used by the two guidance cases. Its one-based target vector is

$$[1, 2, 3, 4, 5, 6, 7, 8, 1, 2, 3, 4, 5, 6, 7, 8, 1, 2, 3, 4],$$

so targets 1–4 receive three defenders and targets 5–8 receive two. A direct reset audit confirms that the same vector is stored in the three-dimensional case preset and consumed without remapping by the guidance environment. This verifies the data interface between the stored assignment and guidance stages. It does not claim that the representative assignment produced by the separate 30-seed reviewer study is identical to this legacy case artifact.

D. Representative Three-Dimensional Engagements

R-MAPPO and an $N = 4$ PN reference are evaluated from the audited initial states. PN uses group-average time-to-go for longitudinal coordination and the same axial and normal load limits. Figure 7 presents the saved representative trajectories.

All saved impacts satisfy the 3-m criterion. The maximum R-MAPPO command magnitudes $[|n_x|, |n_y|, |n_z|]$ are $[0.9882, 0.9975, 0.6061]$ in Case 1 and $[1.0000, 1.0000, 0.7749]$ in Case 2. Both methods succeed in these selected episodes; therefore, Table III supports feasibility and describes synchronization, not statistical superiority or a success-rate difference. Method-to-method interception times are not compared because the two archived evaluators aggregate group time differently.

E. Delay, Noise, Packet Loss, and Failure Case

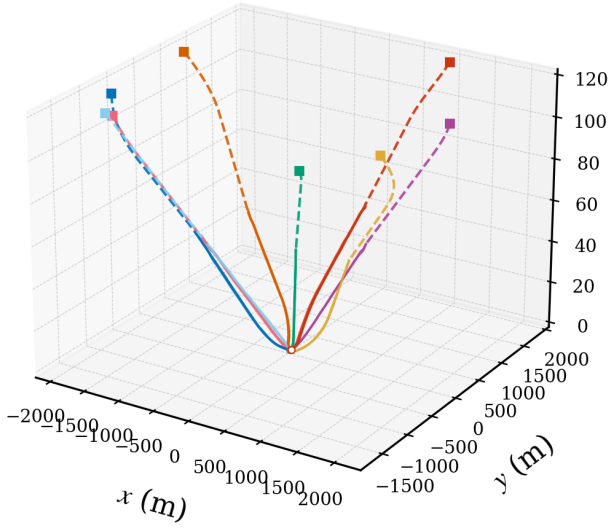
The final trajectories above contain no injected communication impairment. We therefore report two separate tests and keep their criteria distinct. First, one illustrative mathematical-simulation run per case applies a 50-ms sensing delay, 3-m position-noise standard deviation, 0.3-m/s velocity-noise standard deviation, and a 0.30-s first-order command lag. Both cases record 20/20 hits and 8/8 synchronized groups with a 0.25-s maximum spread. This is a selected run, not a robust success-rate estimate.

Second, a software channel-emulation test applies a 100-ms observation delay, 50-ms action delay, normalized observation/action noise standard deviations of 0.003/0.02, and 1% command dropout to five policy endpoints. It uses randomized geometry, a 12-m hit radius, and a 3-s synchronization threshold, so its rates are not directly comparable with Table III.

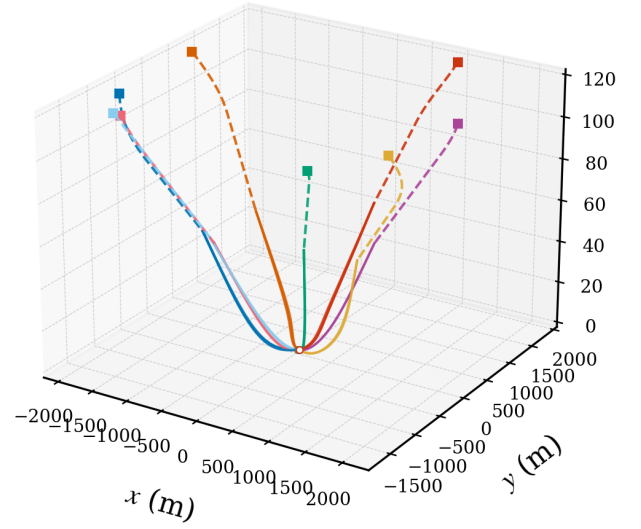
The full-history and GRU-reset rows use identical seeds and the same recurrent checkpoint; the reset diagnostic zeros its hidden state before every action. Resetting preserves all-hit outcomes, but Case-2 episode 7 synchronizes only seven of eight groups and reaches a 39.55-s worst spread. This inference-only intervention supports the relevance of temporal state in that executed case. It is not a retrained feed-forward baseline or a multi-seed architectural ablation, so no independent causal performance gain is claimed for the GRU.

The Case-1 impaired episode 6 is a delayed-coordination failure: all 20 interceptors reach their targets, but only seven of eight groups meet the 3-s criterion and the worst group spread is 43.65 s. This distinction matters operationally because individual hit success does not imply coordinated success. The endpoint objects ran in one Python process; the experiment is software channel emulation, not hardware-in-the-loop or semi-physical validation.

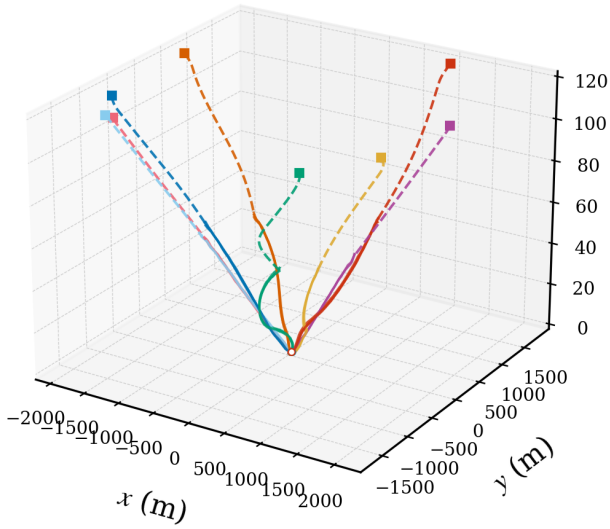
Partial interceptor loss, stuck actuators, and held-out maneuver families were not executed under the final 3-m cases. The present evidence therefore does not establish failure-reassignment robustness or out-of-distribution generalization. Real deployment additionally requires timestamped network transport, estimator bias/drift models, processor latency and jitter measurements, actuator dynamics, collision avoidance, terrain/weather effects, and flight or genuine HIL tests.



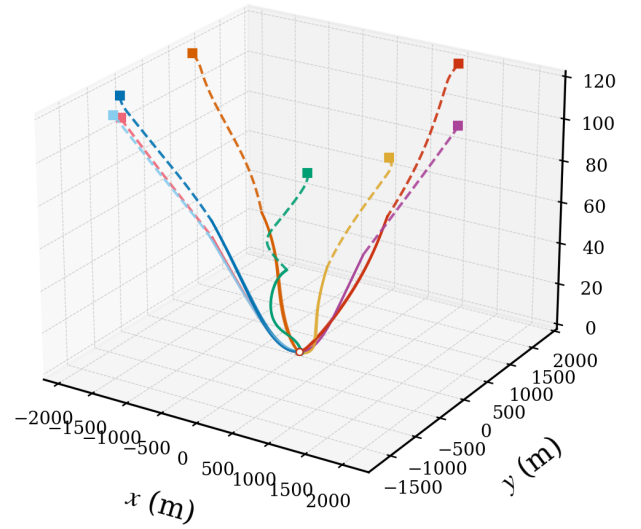
(a) R-MAPPO, Case 1



(b) PN, Case 1



(c) R-MAPPO, Case 2



(d) PN, Case 2

Fig. 7. Representative three-dimensional engagements for the piecewise evasive (Case 1) and sinusoidal weaving (Case 2) targets. Solid and dashed segments show defender and target motion, respectively.

TABLE III
REPRESENTATIVE-RUN THREE-DIMENSIONAL RESULTS ($n = 1$ PER METHOD/CASE)

Method	Case	Interceptor hits	Groups hit	Groups synchronized	Mean group spread (s)	Maximum spread (s)
R-MAPPO	1	20/20	8/8	8/8	0.1250	0.2500
PN ($N = 4$)	1	20/20	8/8	8/8	0.1563	0.4000
R-MAPPO	2	20/20	8/8	8/8	0.1688	0.3000
PN ($N = 4$)	2	20/20	8/8	8/8	0.1438	0.3500

VI. CONCLUSION

This paper formulated capacity-constrained many-to-one target assignment and three-dimensional cooperative guidance as two auditable stages. The revised IDBO study uses the

same 20×8 survival-probability objective as the manuscript and shows monotone incumbent cost under explicit elitist acceptance, while also showing that the tested coefficient decay does not improve final cost over the fixed setting. The

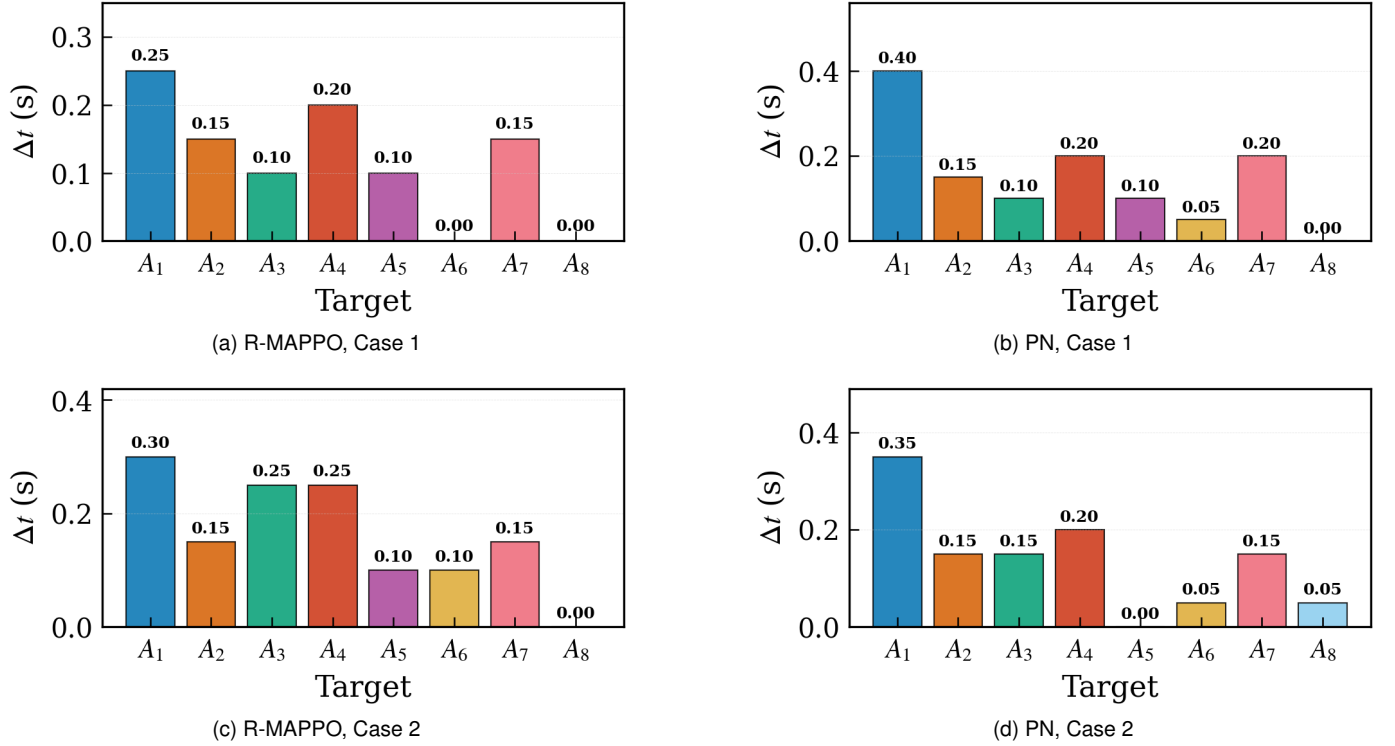


Fig. 8. Within-group terminal arrival-time spreads for the four representative runs. The dashed threshold is 0.5 s.

TABLE IV
SOFTWARE TEMPORAL-STATE AND CHANNEL DIAGNOSTICS (10 EPISODES)

Case	Diagnostic	All-hit	All-sync	Worst spread (s)
1	Full history	10/10	10/10	0.60
1	GRU reset	10/10	10/10	0.60
1	Channel impaired	10/10	9/10	43.65
2	Full history	10/10	10/10	0.20
2	GRU reset	10/10	9/10	39.55
2	Channel impaired	10/10	10/10	0.20

recurrent MAPPO description now matches the evaluated actor architecture and three-dimensional action space.

In the four representative case runs, both R-MAPPO and capacity-matched PN intercept all targets, with maximum within-group spreads of 0.25/0.30 s and 0.40/0.35 s, respectively. A separate software channel test exposes one delayed coordination failure despite complete individual interception. These results support feasibility under the stated conditions, not global optimizer convergence, policy equilibrium, statistical superiority, or hardware readiness. Future work will require genuine distributed optimization, multi-seed reward and architectural retraining, loss-triggered reassignment, held-out maneuver tests, estimator and actuator models, and processor-in-the-loop, HIL, and flight validation.

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APPENDIX A NOTATION TABLE

Table V summarizes the main symbols used in the target assignment and R-MAPPO formulations.

TABLE V
CONCISE NOTATION SUMMARY

Symbol	Description
$\mathcal{U}, \mathcal{T}$	Interceptor and target sets.
M, N	Numbers of interceptors and targets.
$X_{ij}, \mathbf{X}$	Binary assignment variable and assignment matrix.
P_{ij}	Interception probability weight for pair (u_i, t_j) .
J	Target-assignment objective function.
$L_{\max}$	Maximum number of interceptors assigned to one target.
$\mathbf{x}, x_i$	Discrete assignment vector and target selected by interceptor u_i .
K, P	IDBO iteration budget and population size.
s_k, m_k	Mutation strength and number of modified components at iteration k .
$w_{\text{ZEM}}, w_{\sigma}$	ZEM and angular weights in P_{ij} .
$\mathbf{o}_i^{(t)}$	Local observation of interceptor i at time step t .
s_t	Centralized state used by the critic.
$\mathbf{e}_{i,t}, \mathbf{h}_{i,t}$	Encoded feature and GRU hidden state.
$\mathbf{a}_i = [n_x, n_y, n_z]^{\top}$	Three-dimensional normalized load command.
π_{θ_i}, V_{ϕ}	Decentralized actor policy and centralized value function.
$r_t(\theta), \hat{A}_t$	PPO probability ratio and advantage estimate.
$\gamma_{\text{RL}}, \lambda_{\text{GAE}}, \epsilon$	Discount, GAE, and PPO-clipping parameters.
$r_i^{(t)}$	Reward of interceptor i at time step t .
w_q, α_q	Reward weights and term-specific shaping coefficients.
$d_{\text{hit}}, \tau_{\text{sync}}$	Hit radius and group synchronization threshold.
$\mathcal{G}_i$	Interceptors assigned to the same target as u_i .
$E_{\text{sync},j}, E_{\text{miss}}$	Group arrival spread and mean miss-distance metrics.